



Enhanced Cross-Task EEG Classification: Domain Adaptation with EEGNet

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Abstract. In the domain of electroencephalogram (EEG) research, the generalization of workload classification across different tasks, subjects, and channel configurations remains a significant challenge, primarily due to the field's tradition of conducting within-task studies in controlled, lab-based environments. This study pioneers the application of domain adaptation techniques to EEG data, aiming to transcend these limitations by facilitating consistent cross-task classification of workload levels. Central to our methodology is the integration of source localization techniques that render EEG data channel agnostic, thus enhancing our model's capacity to generalize across diverse channel configurations. By utilizing two disparate datasets—one derived from arithmetic tasks and the other from working memory (n-back) tasks—we implement binary classification to discern between low and high workload states. The core of our approach is a modified Convolutional Neural Network (CNN) model, EEGNet, which is specifically designed to capture the temporal and spatial dynamics inherent in EEG data. Enhanced with Maximum Classifier Discrepancy (MCD) for domain adaptation, and bolstered by source localization, this strategic combination enables an impressive 81.76% accuracy in cross-task classification. The success of our strategy in cross-task, cross-channel, and cross-subject classification not only demonstrates its potential for enhancing the generalizability of EEG data analysis but also marks a significant step forward in applying EEG-based workload classification in real-world scenarios, beyond the confines of laboratory settings. The implications of this research are vast, offering a promising avenue for the generalization of EEG data classification across various domains.

Keywords: EEG · Cognitive Workload · Cross Task · Domain Adaptation

1 Introduction

Cognitive workload, defined as the mental effort required to perform a task, plays a pivotal role in various domains such as aviation, healthcare, and human-computer interaction [1]. Accurate assessment of cognitive workload not only

enhances task performance but also contributes to overall well-being and safety. One promising avenue for quantifying cognitive workload is through the analysis of electroencephalography (EEG) signals, which reflect brain activity associated with different cognitive states. Understanding cognitive processes, such as reasoning and problem-solving, is key to deciphering the brain's operations amid its complex network of neurons and synapses [2]. Despite EEG's acknowledged capability in measuring cognitive workload [3,4], its application faces challenges in consistency across varying tasks and individuals due to the inherent differences in brain dynamics and signal properties. In the realm of automation, the emphasis is placed on developing machines capable of recognizing human cognitive states, aiming to enhance the system's overall effectiveness [5]. EEG data is widely used for monitoring cognitive workload in fields such as neuroscience, psychology, and human-computer interaction. However, the task-specific nature of EEG signals poses a challenge for generalizing workload classifiers across different tasks.

The EEG signals may vary among individuals and tasks, presenting both within-subject and cross subject differences, as well as variations across different tasks. These differences influence the creation of models specific to individual subjects, models applicable across different subjects, and models suitable for cross tasks [6]. Workload classifiers are trained and evaluated on the same task or dataset, limiting their generalizability to new tasks or domains. This limitation hinders the deployment of workload classification systems in real-world settings where tasks may vary widely. Addressing this gap requires the development of robust workload classifiers capable of adapting to diverse task environments—a method known as cross task classification.

To address these challenges, researchers have employed advanced signal processing techniques, feature extraction methods, and machine learning algorithms [7,8]. In particular, domain adaptation techniques have been identified as promising for aligning feature distributions between varied experimental conditions, thereby enhancing classifier robustness [9,10]. Yet, many of these studies focus predominantly on single-task scenarios, leaving a gap in comprehensive analyses that span across multiple tasks, subjects, and channels.

The concept of cross-task modeling involves training a model on one task and then testing it on another task with similarities but distinctions [11–13]. Although different cognitive tasks may trigger different cognitive resources, cognitive workload primarily revolves around the overall amount of cognitive resources utilized, rather than their specific composition. Consequently, there is potential to devise a versatile cognitive workload recognition model capable of discerning workload levels across diverse tasks [14].

Several approaches have been proposed to address the challenge of cross-task and cross-domain EEG classification. Baldwin et al. [15] attempted cross-task workload classification using artificial neural networks, but achieved poor performance below chance levels. Ke et al. [16] had more success using feature selection and support vector machine regression to identify task-independent EEG features for cross-task workload estimation. Zhang et al. [17] proposed a deep

domain adaptation framework using adversarial training to align EEG feature distributions across tasks and subjects. Dimitrakopoulos et al. [18] developed a method using functional connectivity features and sequential feature selection to achieve 87% accuracy for cross-task workload classification. More recently, Zhou et al. [19] employed a deep domain adaptation approach with feature alignment and adversarial training, improving cross-subject workload classification accuracy by 2-9% over baseline methods. Lan et al. [20] applied domain adaptation techniques to EEG-based emotion recognition achieving significant accuracy improvements of up to 13.40% across datasets, showcasing the potential for enhanced subject-independent emotion recognition. These studies demonstrate the potential of domain adaptation techniques to improve the generalization of EEG-based mental state classification across different tasks and subjects.

This paper makes several significant contributions to the advancement of EEG-based classification systems, emphasizing enhanced adaptability and generalizability across various analytical dimensions:

1. **Cross-Task and Cross-Subject Generalizability:** Our framework excels in generalizing EEG classification across diverse cognitive tasks and varied subjects, demonstrating robust performance from Mental Arithmetic to N-Back tasks. This adaptability is powered by our innovative use of domain adaptation techniques and source localization, which ensures effective classification regardless of task or subject variability. Such capabilities are crucial for applications that demand reliable cognitive assessments across different populations.
2. **Effective Domain Adaptation:** The implementation of Maximum Classifier Discrepancy (MCD) for domain adaptation significantly enhances the model's performance in cross-task scenarios. Our findings, including an increase in accuracy to 81.76% under domain adaptation, validate the efficacy of MCD in adjusting the model to handle variations arising from different experimental conditions and task complexities.
3. **Standardization of EEG Data Processing:** By standardizing EEG data into a common brain space using source localization techniques, our study sets a new precedent for ensuring consistent data comparability across sessions and channels. This approach not only aids in maintaining the integrity of EEG analyses but also supports the reproducibility of research findings across varied setups and equipment.

The remaining sections of the paper are structured as follows. Section 2 provides a brief overview of datasets, EEG pre-processing, Source Localization, Model Architecture, domain adaptation and relevant methodologies. The ensuing section, Section 3 delves into the key discoveries derived from the analysis and contrasts the outcomes to assess the effectiveness of the proposed approaches. Section 4 offers concluding remarks for the entire paper.

2 Methods

Utilizing datasets from varied cognitive tasks, the methods involve thorough preprocessing, source localization techniques, and the deployment of advanced models for classification. These steps aim to assess the generalizability and efficacy of EEG signal classification across different cognitive states.

2.1 Dataset

To explore the robustness of EEG classification across various conditions, our study leverages two distinct datasets corresponding to different cognitive tasks. The first dataset comprises EEG recordings from participants engaged in a Mental Arithmetic task and is sourced from the study by Zyma et al. [21], which is publicly available on the PhysioNet platform [22]. The second dataset is derived from a Working Memory task (n-back), obtained from the research conducted by Shin et al. [23]. These datasets provide a foundation for evaluating the generalizability of EEG classification across tasks, channels, and subjects, presenting a comprehensive perspective on the applicability of our models to varied cognitive evaluations.

Dataset 1: Mental Arithmetic Task The Mental Arithmetic task dataset in this study consists of EEG recordings from 36 healthy participants, using a Neurocom monopolar 23-channel EEG system following the 10-20 electrode placement. The EEG data, sampled at 500 Hz, include a 180-second resting phase and a 60-second active mental arithmetic phase. For balanced analysis, only the central 60 seconds of the resting phase were used, equivalent to the active task duration. We categorize the resting phase as representing low workload and the active phase as high workload conditions.

Dataset 2: Working Memory (n-back) Task The N-Back task dataset was collected from 26 healthy individuals, utilizing a 30 channel Brain Products BrainAmp EEG with a sampling frequency of 200 Hz, placed according to the international 10-5 system. The dataset included three task conditions: 0-back, 2-back, and 3-back, designed to assess varying cognitive loads. For our analysis, we specifically focused on the 0-back and 3-back tasks to represent low and high workloads, respectively.

2.2 EEG Data Processing

This section details the precise EEG data processing steps implemented to standardize and enhance the signals.

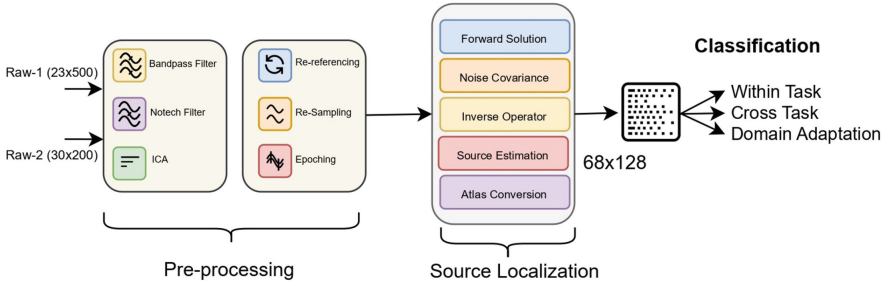


Fig. 1. Signal Processing and Classification Pipeline: Data cleaning and pre-processing with filtering and ICA, followed by source localization to make datasets channel-agnostic and classification through within-task, cross-task, and domain adaptation strategies.

Preprocessing To ensure the integrity and comparability of EEG data across different sessions and conditions, the following preprocessing steps are employed (shown in Figure 1): bandpass filtering to restrict the frequency range to 0.5–45 Hz, notch filtering at 50 Hz to remove electrical line noise, artifact correction using Independent Component Analysis (ICA) to remove eye blinks and cardiac signals, re-referencing the data to an average reference, resampling to 128 Hz to standardize sampling frequency and reduce computational load, and data epoching where fixed-length segments of 1 second with a 0.5-second overlap are created for continuous data coverage during analysis.

Source Localization Following the preprocessing of EEG data, source localization techniques are applied to convert data from electrode space to a standardized brain space using the Desikan-Killiany atlas. This step is essential for comparing and combining datasets effectively.

The process involves setting up the source space and boundary element model (BEM) using a standardized MRI subject (‘fsaverage’) from MNE’s configuration. Specific steps include creating a source space with octahedron subdivisions (oct6), generating the BEM model and solution, and defining the transformation applicable to ‘fsaverage’.

The localization to the source space is achieved through the following steps (as shown in Figure 1):

- **Forward Solution:** Compute a forward solution [24] to model the relationship between the source spaces and the EEG sensors,

$$\mathbf{F} = \mathbf{G}\mathbf{S},$$

where $\mathbf{G}$ is the lead field matrix representing the gain of each sensor to each source and $\mathbf{S}$ denotes the source strengths.

- **Noise Covariance:** Estimate the noise covariance [25] from the data to regularize the source estimation,

$$\mathbf{C} = \frac{1}{N-1} \sum_{i=1}^N (\mathbf{E}_i - \bar{\mathbf{E}})(\mathbf{E}_i - \bar{\mathbf{E}})^\top,$$

where $\mathbf{E}_i$ are the EEG data epochs and $\bar{\mathbf{E}}$ is the mean over all epochs.

- **Inverse Operator:** Create an inverse operator using the forward solution and noise covariance. This operator is used to compute the neural source estimates,

$$\mathbf{M} = (\mathbf{G}^\top \mathbf{C}^{-1} \mathbf{G} + \alpha \mathbf{I})^{-1} \mathbf{G}^\top \mathbf{C}^{-1},$$

where α is the regularization parameter, and $\mathbf{I}$ is the identity matrix.

- **Source Estimation:** Apply the inverse operator to the epochs to retrieve source time courses, which are then mapped to the Desikan-Killiany atlas [26],

$$\mathbf{S}_t = \mathbf{M}\mathbf{E},$$

and is a crucial step for determining the localization of neural activity [27].

- **Atlas Conversion:** Extract the mean time course from each brain region as defined by the atlas to facilitate direct comparisons across subjects and sessions,

$$\mathbf{T}_r = \frac{1}{N_r} \sum_{j \in R} (\mathbf{S}_{t,j}),$$

where R represents the set of sources within the region r and N_r is the number of sources in r .

This methodology allows the EEG data to be represented in a common brain space, enhancing the interpretability and comparability of the results across different datasets. Following this step, the spatial dimension of the data corresponds to 68 channels, representing the 68 brain regions defined in the Desikan-Killiany atlas. Consequently, the shape of each epoch is transformed to 68x128, indicating 68 brain regions and 128 samples per second. Figure 2 shows the output after EEG source localization which is used as the input for the classification.

2.3 Classification Methods

This section outlines the advanced classification techniques used to analyze EEG data, focusing on the models and configurations designed specifically for identifying cross-task workload levels.

Base Model (Modified EEGNet) EEGNet [9] is selected as the base model for classification due to its compact and efficient architecture specifically designed to handle the high-dimensional and noisy nature of EEG signals. This modified version of EEGNet (as shown in Table 1) is particularly tailored to align with the specific requirements of our EEG datasets, which consist of

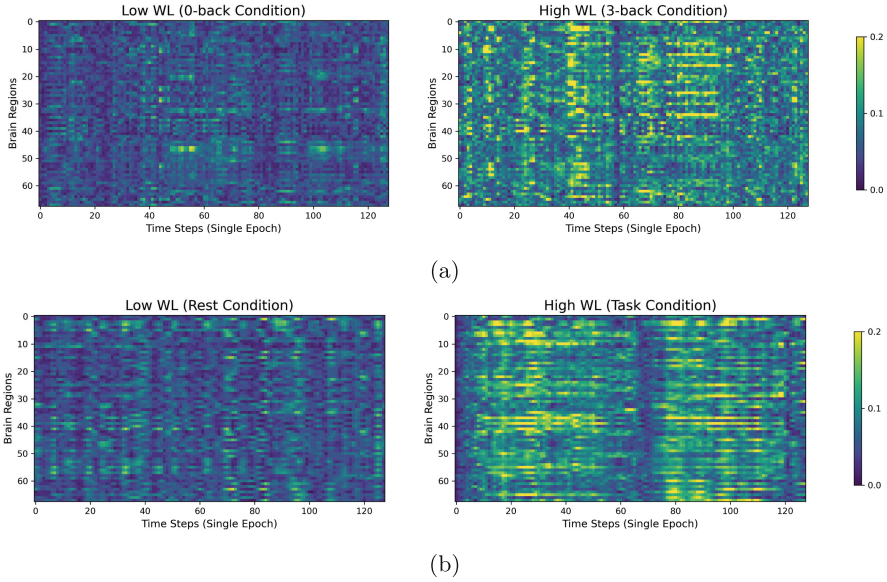


Fig. 2. Output after EEG Source Localization Visualization (a) Arithmetic task dataset (b) Working memory (n-back) task dataset

data segmented into one-second epochs, each comprising 128 time points and originating from 68 distinct brain regions or channels.

The architecture has been modified to include:

- **Temporal and Spatial Filtering:** The model uses $F1$, the number of temporal filters, to initially transform the EEG data across the time dimension. This is complemented by a depth multiplier, D , representing the number of spatial filters per channel, which helps in extracting spatial features specific to each EEG channel through depthwise convolutions.
- **Complex Feature Extraction:** The number of pointwise filters, $F2(= D \times F1)$, is optimized to further process the features, enhancing the model's ability to distinguish between different brain states or conditions effectively.
- **Classification:** The output layer is designed to reflect the number of classes, N , which categorizes the EEG data into predefined classes based on the learned features.

This configuration not only ensures that EEGNet remains lightweight and fast but also improves its accuracy and generalizability across different EEG tasks. Multiple configurations of $F1$ and D , such as (4, 2) and (8, 2), were tested. The best results were achieved with $F1 = 8$ and $D = 4$, which we have used throughout the study to optimize performance, achieving a balance between computational efficiency and the capability to perform detailed and robust classification of EEG signals.

Domain Adaptation using Maximum Classifier Discrepancy In our approach, we employ the Maximum Classifier Discrepancy (MCD) [28] technique for domain adaptation across two distinct EEG tasks: an arithmetic task and a working memory task. MCD leverages adversarial learning principles to align feature distributions from the source and target domains by maximizing the discrepancy between two classifiers, $F1$ and $F2$, both instances of a modified EEGNet model. The feature generator G transforms EEG signals, ensuring their dimensions are suitable for EEGNet classification. The alignment between domains is driven by the discrepancy loss function $d(p_1, p_2)$, which quantifies the mean absolute error between the classifiers' predictions, serving as a critical component of the adversarial training mechanism integral to MCD.

Table 1. Modified EEGNet architecture, where C represents the number of channels or brain regions, set at 68. T denotes the number of time points per epoch, which is 128 in a one-second sample. $F1$ is the number of temporal filters, D the depth multiplier indicating the number of spatial filters, $F2(= D \times F1)$ the number of pointwise filters, and N the number of classes.

Layer	Filters	Size	Output	Activation	Options
Reshape	-	-	$C \times T \times 1$	-	-
Conv2D	$F1$	$(1, \frac{T}{2})$	$C \times T \times F1$	Linear	Padding=same
Batch Normalization	-	-	$C \times T \times F1$	-	-
Depthwise Conv2D	$D \times F1$	$(C, 1)$	$1 \times T \times D \times F1$	Linear	Padding=valid, Depth=D, MaxNorm(1)
Batch Normalization	-	-	$1 \times T \times D \times F1$	-	-
Activation	-	-	$1 \times T \times D \times F1$	ELU	-
Average Pooling2D	-	$(1, 4)$	$1 \times \frac{T}{4} \times D \times F1$	-	-
Dropout	-	-	$1 \times \frac{T}{4} \times D \times F1$	-	Rate=0.25
Separable Conv2D	$F2$	$(1, \frac{T}{8})$	$1 \times \frac{T}{8} \times F2$	Linear	Padding=same
Batch Normalization	-	-	$1 \times \frac{T}{8} \times F2$	-	-
Activation	-	-	$1 \times \frac{T}{8} \times F2$	ELU	-
Average Pooling2D	-	$(1, 8)$	$1 \times \frac{T}{32} \times F2$	-	-
Dropout	-	-	$1 \times \frac{T}{32} \times F2$	-	Rate=0.25
Flatten	-	-	$F2 \times \frac{T}{32}$	-	-
Dense	-	-	$N \times (F2 \times \frac{T}{32})$	ReLU	MaxNorm(0.25)
Dense	-	-	1	Sigmoid	MaxNorm(0.25)

Generator Architecture: The generator network is expertly constructed to transform EEG input while preserving its original shape for compatibility with EEGNet classifiers. It is a sequential convolutional model that ensures the dimensions of the input EEG data, consisting of 68 channels by 128 time points, remain intact through each layer. The use of ‘same’ padding and specific activation functions within the network facilitates this preservation of spatial dimensions. The detailed architecture of the generator network, including the layers, filter sizes, and activations, is delineated in Table 2, highlighting its role as an integral component in our domain adaptation framework.

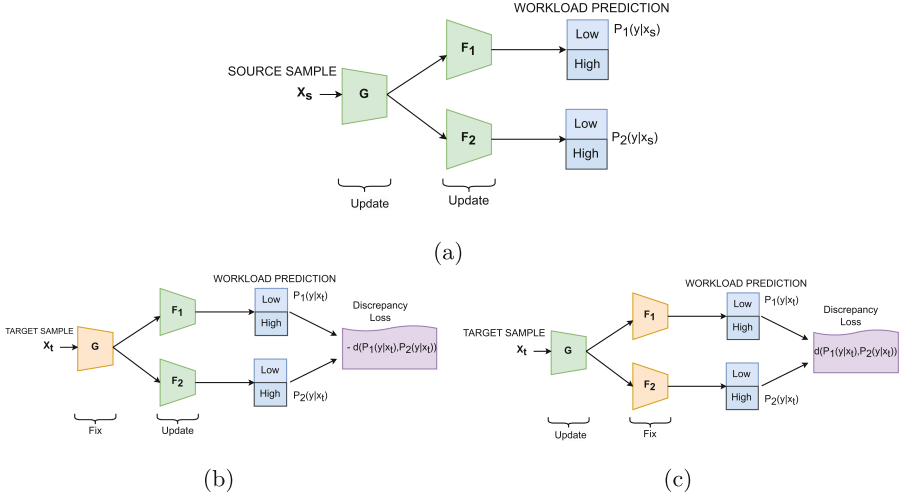


Fig. 3. Schematic of the adaptive workload prediction model: (a) Training with source samples X_s , utilizing a shared generator G and two distinct classifiers F_1 and F_2 for workload categorization. (b) showcases the maximization of the discrepancy on the target sample X_t , fixing the generator G and updating classifier F_1 and F_2 . (c) illustrates the minimization of discrepancy, updating G and fixing F_1 and F_2 , to refine the prediction accuracy on X_t .

Table 2. Generator Network Architecture

Layer	Filters	Size	Output	Activation	Options
InputLayer	-	-	$68 \times 128 \times 1$	-	-
Conv2D	32	(3, 3)	$68 \times 128 \times 32$	ReLU	Padding=same
Batch Normalization	-	-	$68 \times 128 \times 32$	-	-
Conv2D	64	(3, 3)	$68 \times 128 \times 64$	ReLU	Padding=same
Batch Normalization	-	-	$68 \times 128 \times 64$	-	-
Conv2D	1	(3, 3)	$68 \times 128 \times 1$	Sigmoid	Padding=same

Discrepancy Loss Function: A key component of our MCD approach is the discrepancy loss function, which quantifies the difference in the classifiers' probabilistic outputs. The discrepancy loss function is defined as the mean absolute error (MAE) between the output probabilities of the two classifiers:

$$d(p_1, p_2) = \frac{1}{N} \sum_{i=1}^N |p_1(Y|X_t^{(i)}) - p_2(Y|X_t^{(i)})|,$$

where $p_1(Y|X_t^{(i)})$ and $p_2(Y|X_t^{(i)})$ are the predicted probabilities by classifiers F_1 and F_2 for the target sample $X_t^{(i)}$, and N is the number of target samples.

Training Objectives: The training involves a three-step process (as shown in Figure 3), each targeting a specific objective:

1. **Step A (Source Domain Training):** Train the classifiers and generator on the source domain data to minimize the binary cross-entropy loss:

$$\min_{G, F1, F2} \mathcal{L}_{\text{CE}}(X_s, Y_s),$$

$$\mathcal{L}_{\text{CE}}(X_s, Y_s) = -\frac{1}{N} \sum_{i=1}^N \left[Y_s^{(i)} \log(p(Y_s^{(i)} | X_s^{(i)})) + (1 - Y_s^{(i)}) \log(1 - p(Y_s^{(i)} | X_s^{(i)})) \right].$$

2. **Step B (Discrepancy Maximization on Target Data):** Adapt the classifiers to maximize the discrepancy on the target domain data while retaining performance on the source domain:

$$\min_{F1, F2} \mathcal{L}_{\text{CE}}(X_s, Y_s) - \mathcal{L}_{\text{adv}}(X_t),$$

$$\mathcal{L}_{\text{adv}}(X_t) = \mathbb{E}_{X_t \sim X_t} [d(p1(y|X_t), p2(y|X_t))],$$

with the discrepancy d measured as the mean absolute error between the probabilistic outputs of the classifiers.

3. **Step C (Minimization of Discrepancy on Target Data):** Optimize the generator to minimize the discrepancy between the classifiers' predictions on the target data:

$$\min_G \mathcal{L}_{\text{adv}}(X_t).$$

The hyperparameter n determines the iteration count for each mini-batch within the training cycle. It serves as a tuning parameter in the original MCD framework, calibrating the equilibrium between the generator's and the classifiers' objectives during the adversarial training process.

This training regimen is iterated to refine the model's ability to classify EEG data correctly, irrespective of the task. The loss functions are tailored to ensure that the adversarial nature of MCD leads to domain-invariant feature extraction, enhancing the model's generalization capabilities.

2.4 Classification Experiments

In order to evaluate the efficacy of our domain adaptation approach using Maximum Classifier Discrepancy (MCD), we conducted a series of classification experiments. These experiments were designed to assess the performance of our model under various conditions and are divided into three main cases:

Case 1: Within-Task Classification The first case investigates the performance of our model when training and testing are conducted within the same task domain. This serves as a baseline to understand the model's capability in a controlled setting, where the feature distribution between training and testing sets is consistent.

Case 2: Cross-Task Classification without Domain Adaptation In Case 2, we assess the cross-task classification capabilities of our approach without the aid of domain adaptation techniques. We conduct two sets of experiments to evaluate the baseline generalizability of the network: first, by training on the dataset from the arithmetic task (D1) and testing on the working memory task (D2), and second, by reversing this approach—training on the working memory task (D2) and testing on the arithmetic task (D1). These assessments provide insight into the inherent ability of the network to transfer learning between different cognitive tasks without domain-specific tuning.

Case 3: Cross-Task Classification with Domain Adaptation In the third case, our focus shifts to evaluating the effectiveness of the MCD technique with the three-step training process for domain adaptation across tasks. We modulate the domain adaptation intensity by varying the hyperparameter n – denoting the number of iterations per mini-batch – with values of 2, 3, and 4. This variation allows us to observe the nuanced impacts of domain adaptation on knowledge transfer between the arithmetic and working memory tasks. For each setting, the domain adaptation procedure is conducted twice, with each dataset serving as the source and then as the target, ensuring a comprehensive evaluation of the model’s adaptability to both tasks.

3 Results

In an effort to substantiate the performance of our EEG classification framework, we executed a comprehensive set of experiments. Both the Mental Arithmetic and Working Memory datasets were divided into training and testing sets with a 70-30 split. Our experiments were conducted over 50 epochs with an early stopping mechanism set at 5 epochs to prevent overfitting. We selected a batch size of 512 to optimize the computational efficiency and convergence rate. The optimization of our model was conducted using the Adam optimizer, known for its efficacy in handling sparse gradients on noisy problems. All experiments were implemented using TensorFlow, which provides a robust backend framework conducive to extensive neural network experimentation.

The results from these experiments, as detailed in Table 3, highlight the model’s capability to not only adapt within specific tasks but also to generalize across different tasks, subjects, and channels—key factors for the deployment of EEG classification systems in real-world applications.

Within-Task Classification Results Within-task classification revealed a superior performance, with an accuracy of 85.23% for dataset D1. This high accuracy, coupled with an F1 score of 84.57%, suggests that the model reliably differentiates cognitive states within the same type of task. The close precision and recall scores indicate a balanced classification capability, with dataset D2 showing only a slight reduction in performance across all metrics. The success

Table 3. Classification Performance Metrics Across Cases and Conditions

Case	Condition	Accuracy	Precision	Recall	F1 Score
Within-Task	D1	85.23	84.76	83.68	84.57
	D2	84.67	83.59	82.45	83.54
2cmCross-Task without DA	D1 to D2	65.12	67.34	64.28	65.75
	D2 to D1	62.89	60.73	63.54	61.67
Cross-Task with DA	D1 to D2, $n = 2$	75.68	74.56	73.47	74.32
	D2 to D1, $n = 2$	74.21	73.89	72.58	73.45
	D1 to D2, $n = 3$	80.47	81.02	79.56	80.29
	D2 to D1, $n = 3$	79.34	78.69	77.85	78.27
	D1 to D2, $n = 4$	81.76	81.49	81.37	80.53
	D2 to D1, $n = 4$	80.42	79.67	80.58	80.36

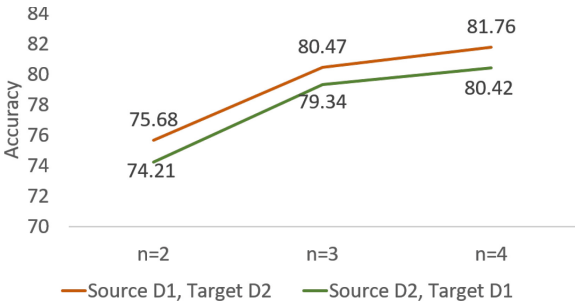


Fig. 4. Graph displays the performance trends of two domain adaptation scenarios with training iterations ($n=2$, $n=3$, $n=4$).

in this case underscores the model’s proficiency in discerning between low and high workload conditions when the task remains constant.

Cross-Task Classification without Domain Adaptation Transitioning to cross-task classification without domain adaptation presented a notable challenge for the model, as evidenced by a significant decrease in accuracy, with the highest being 65.12% when training on D1 and testing on D2. This dip reflects the inherent difficulty in transferring knowledge between different cognitive tasks without additional adaptation mechanisms. Although precision remained relatively stable, a drop in recall for D2 when tested on D1 suggests a propensity for the model to miss true positive instances under a new domain, indicating the necessity for domain-specific tuning to improve generalizability.

Cross-Task Classification with Domain Adaptation Introducing domain adaptation markedly enhanced the model’s adaptability across tasks. Notably, the accuracy improved to 81.76% for D1 tested on D2 with $n = 4$ adaptation

iterations, emphasizing the significant role of domain adaptation in cross-task classification scenarios. This represents a substantial improvement from the non-adapted models and demonstrates that the model can be effectively tuned for different cognitive tasks using the MCD approach. The increment in the number of adaptation iterations (n) from 2 to 4 yielded consistent improvements across all performance metrics as shown in Figure 4, suggesting that a more intensive domain adaptation process is beneficial for model robustness. The high F1 score for $n = 4$, 80.53%, further reinforces the model's precision and recall harmony when the domain adaptation is optimally configured.

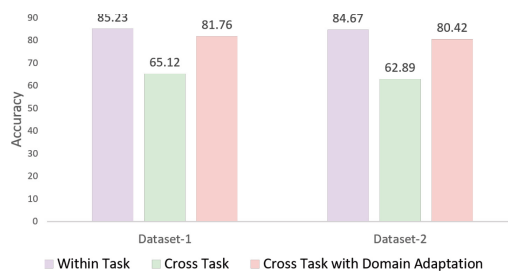


Fig. 5. Comparison of Classification Accuracies: illustrating the performance of both datasets for within task, cross-task with and without domain adaptation.

The results delineate a clear progression from within-task competency to effective cross-task classification with the integration of domain adaptation techniques (depicted in Figure 5). The gradual enhancement of performance metrics with increased domain adaptation efforts highlights the importance of this approach in extending the scope across varied cognitive evaluations.

The results of this study hold substantial implications for the advancement of EEG-based classification systems, particularly in their application across different tasks, subjects, and channels. The robust within-task performance sets a standard for the model's capabilities and provides a reference point for assessing cross-task generalizability. The differential success rates in cross-task classification underscore the challenges and reveal opportunities for enhancement through domain adaptation. Notably, the improvement in accuracy and F1 scores in the presence of domain adaptation underscores its critical role in surmounting variabilities inherent to cross-task, cross-subject, and cross-channel scenarios. This adaptability is crucial for the practical deployment of EEG classifiers in dynamic real-world settings, where the ability to interpret neural signals consistently, despite variations in task context, subject responses, and channel configurations, is essential. The positive outcomes observed with domain adaptation not only highlight its necessity but also affirm its efficacy in promoting the widespread applicability of EEG classification models.

4 Conclusion

This study has made significant strides in the field of EEG-based classification systems, demonstrating robust capabilities in handling variability across different tasks, subjects, and channels. Through the application of advanced preprocessing techniques, domain adaptation methods, and innovative neural network architectures, we have shown that it is possible to significantly enhance the adaptability and generalizability of EEG classifiers. The key achievements of this study include the successful generalization of the EEG classification model across varied cognitive tasks and among diverse participant groups. By employing the Maximum Classifier Discrepancy (MCD) technique, we improved the model's accuracy in cross-task scenarios, which is a substantial step forward from traditional EEG classifiers that are often limited to specific task conditions. Our approach has proven particularly effective, with notable improvements in classification accuracy reaching up to 81.76% in complex domain adaptation settings. Additionally, the standardization of EEG data through source localization to a common brain space has enhanced the comparability and consistency of data analysis across studies and setups. The implications of our findings are profound, extending beyond the immediate scope of EEG classification. The ability to accurately classify EEG data across varying conditions suggests potential applications in personalized medicine, where EEG-based diagnostics and treatments could be tailored to individual physiological and cognitive profiles. Furthermore, our methodological innovations provide a framework that can be adapted to other modalities of biomedical data, potentially transforming approaches to neural monitoring and intervention. While our results are promising, they also highlight areas for further research. One limitation of the current study is the focus on controlled laboratory tasks. Future work could extend these methods to more naturalistic settings to better capture the complexities of real-world cognitive states. Another limitation is the focus on a binary classification task; future work will extend our approach to multiclass problems to evaluate the performance and robustness of MCD in more complex classification scenarios. Further research should also explore the scalability of the proposed models in larger, more heterogeneous populations to validate the generalizability of the findings. Moreover, exploring the impact of varying the hyperparameters, such as the number of adaptation iterations (n), could provide deeper insights into the dynamics of domain adaptation in neural networks.

In conclusion, this study not only advances our understanding of EEG signal classification but also sets a benchmark for future research in this rapidly evolving field. By addressing both methodological challenges and practical applications, the contributions of this research pave the way for the next generation of EEG classification systems that are more adaptable, reliable, and widely applicable in clinical and research settings. The potential of these systems to revolutionize how we understand and interact with the human brain is immense, promising significant advancements in neuroscience, medicine, and beyond.

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